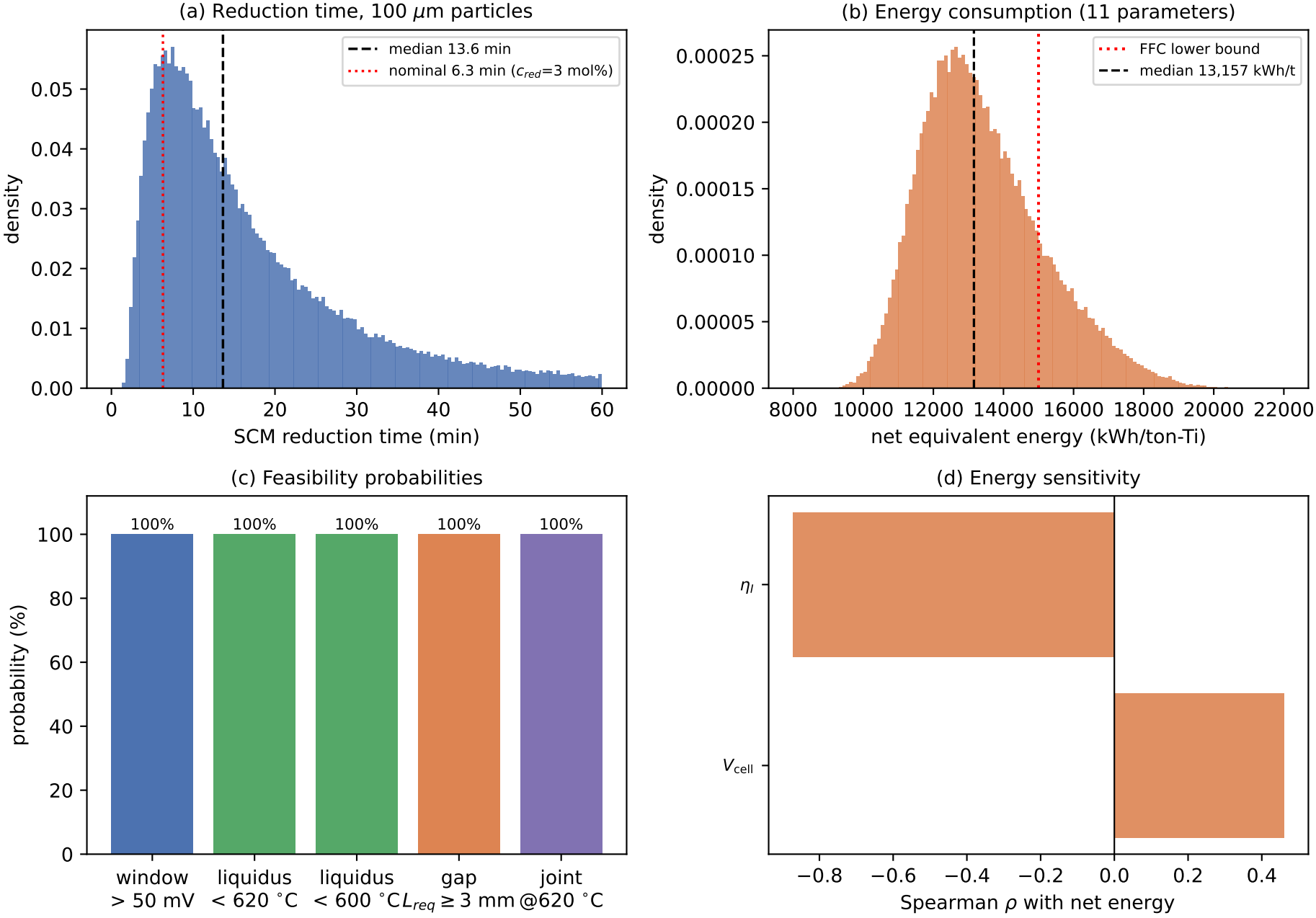
(a) Reduction time, 100 μm particles
median 13.6 min
nominal 6.3 min (c_{red}=3 mol%)
density
SCM reduction time (min)
(b) Energy consumption (11 parameters)
FFC lower bound
median 13,157 kWh/t
density
net equivalent energy (kWh/ton-Ti)
(c) Feasibility probabilities
100%
probability (%)
window > 50 mV
liquidus < 620 ˚C
liquidus < 600 ˚C
gap L_{req} ≥ 3 mm
joint @620 ˚C
(d) Energy sensitivity
η_I
V_{cell}
Spearman ρ with net energy